

# Circular Economy Ontology Network (CEON) - Actor Module

## Metadata

### IRI

<http://w3id.org/CEON/ontology/actor/>

### Title

Circular Economy Ontology Network (CEON) - Actor Module

### Creator

Eva Blomqvist

### Contributor

Huanyu Li

Mikael Lindecrantz

Robin Keskisärkkä

### Date Created

2023-03-16

### License

<https://github.com/LiUSemWeb/CEON/blob/master/LICENSE>

### Version Iri

<http://w3id.org/CEON/ontology/actor/0.1/>

### Version Info

0.1

### Preferred Namespace Prefix

actor

### Preferred Namespace Uri

<http://w3id.org/CEON/ontology/actor/>

### Description

The Actor module of CEON (Circular Economy Ontology Network).

### Covers Requirements

In addition to requirements covered by imported ODPs, covers the following requirements from Onto-DESIDE D3.1: C0-1, C0-2, C3-4, T3-4

## Classes

### Actor Cvn Role<sup>c</sup>

#### IRI

<http://w3id.org/CEON/ontology/actor/ActorCVNRole>

#### Sub Class Of

[ActorCollaborationRole<sup>c</sup>](#)

#### Named Individuals

[collector<sup>ni</sup>](#)

[dismantler<sup>ni</sup>](#)

[manufacturer<sup>ni</sup>](#)

[recycler<sup>ni</sup>](#)

[reseller<sup>ni</sup>](#)

[seller<sup>ni</sup>](#)

[supplier<sup>ni</sup>](#)

[user<sup>ni</sup>](#)

## Actor Collaboration Role<sup>C</sup>

|                              |                                                                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                   | <a href="http://w3id.org/CE0N/ontology/actor/ActorCollaborationRole">http://w3id.org/CE0N/ontology/actor/ActorCollaborationRole</a> |
| <b><u>Sub Class Of</u></b>   | <a href="#">actorODP:Role<sup>C</sup></a>                                                                                           |
| <b><u>Super Class Of</u></b> | <a href="#">ActorCVNRole<sup>C</sup></a><br><a href="#">ActorProcessRole<sup>C</sup></a>                                            |

## Actor Process Role<sup>C</sup>

|                            |                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/actor/ActorProcessRole">http://w3id.org/CE0N/ontology/actor/ActorProcessRole</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">ActorCollaborationRole<sup>C</sup></a>                                                                      |

## Actor Resource Role<sup>C</sup>

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                      | <a href="http://w3id.org/CE0N/ontology/actor/ActorResourceRole">http://w3id.org/CE0N/ontology/actor/ActorResourceRole</a>                                                                                                                                                                                                                                                                                                                                                               |
| <b><u>Sub Class Of</u></b>      | <a href="#">actorODP:Role<sup>C</sup></a>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b><u>Named Individuals</u></b> | <a href="#">buyer<sup>ni</sup></a><br><a href="#">consumer<sup>ni</sup></a><br><a href="#">issuer<sup>ni</sup></a><br><a href="#">owner<sup>ni</sup></a><br><a href="#">producer<sup>ni</sup></a><br><a href="#">provider<sup>ni</sup></a><br><a href="#">reseller<sup>ni</sup></a><br><a href="#">seller<sup>ni</sup></a><br><a href="#">supplier<sup>ni</sup></a><br><a href="#">updater<sup>ni</sup></a><br><a href="#">user<sup>ni</sup></a><br><a href="#">viewer<sup>ni</sup></a> |

## Buying Resource<sup>C</sup>

|                               |                                                                                                                                                                |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                    | <a href="http://w3id.org/CE0N/ontology/actor/BuyingResource">http://w3id.org/CE0N/ontology/actor/BuyingResource</a>                                            |
| <b><u>Sub Class Of</u></b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                          |
| <b><u>Equivalentclass</u></b> | <a href="#">actorODP:ResourceRelation<sup>C</sup></a> <i>and</i> <a href="#">actorODP:participantRole<sup>op</sup></a> value <a href="#">buyer<sup>C</sup></a> |

## Cvn Participation<sup>C</sup>

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/actor/CVNParticipation">http://w3id.org/CE0N/ontology/actor/CVNParticipation</a>                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b><u>Sub Class Of</u></b> | <a href="#">actorODP:CollaborationParticipation<sup>C</sup></a><br><a href="#">actorODP:participatingActor<sup>op</sup></a> some <a href="#">actorODP:Actor<sup>C</sup></a> <i>and</i><br><a href="#">actorODP:participantRole<sup>op</sup></a> some <a href="#">ActorCVNRole<sup>C</sup></a> <i>and</i> <a href="#">actorODP:participationIn<sup>op</sup></a> some <a href="http://w3id.org/CEON/ontology/cvn/CVN">http://w3id.org/CEON/ontology/cvn/CVN<sup>C</sup></a> <i>and</i><br><a href="#">actorODP:CollaborationParticipation<sup>C</sup></a> |

## Consuming Resource<sup>C</sup>

|                        |                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/ConsumingResource">http://w3id.org/CE0N/ontology/actor/ConsumingResource</a>                                  |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                      |
| <b>Equivalentclass</b> | <a href="#">actorODP:participantRole<sup>OP</sup></a> value <a href="#">consumer<sup>C</sup></a> and <a href="#">actorODP:ResourceRelation<sup>C</sup></a> |

## Issuing Resource<sup>C</sup>

|                        |                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/IssuingResource">http://w3id.org/CE0N/ontology/actor/IssuingResource</a>                                    |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                    |
| <b>Equivalentclass</b> | <a href="#">actorODP:participantRole<sup>OP</sup></a> value <a href="#">issuer<sup>C</sup></a> and <a href="#">actorODP:ResourceRelation<sup>C</sup></a> |

## Owning Resource<sup>C</sup>

|                        |                                                                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/OwningResource">http://w3id.org/CE0N/ontology/actor/OwningResource</a>                                     |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                   |
| <b>Equivalentclass</b> | <a href="#">actorODP:participantRole<sup>OP</sup></a> value <a href="#">owner<sup>C</sup></a> and <a href="#">actorODP:ResourceRelation<sup>C</sup></a> |

## Process Participation<sup>C</sup>

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>          | <a href="http://w3id.org/CE0N/ontology/actor/ProcessParticipation">http://w3id.org/CE0N/ontology/actor/ProcessParticipation</a>                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Sub Class Of</b> | <a href="#">actorODP:CollaborationParticipation<sup>C</sup></a><br><a href="#">actorODP:CollaborationParticipation<sup>C</sup></a> and <a href="#">actorODP:participationIn<sup>OP</sup></a> some <a href="#">http://w3id.org/CE0N/ontology/process/Process<sup>C</sup></a> and <a href="#">actorODP:participatingActor<sup>OP</sup></a> some <a href="#">actorODP:Actor<sup>C</sup></a> and <a href="#">actorODP:participantRole<sup>OP</sup></a> some <a href="#">ActorCVNRole<sup>C</sup></a> |

## Producing Resource<sup>C</sup>

|                        |                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/ProducingResource">http://w3id.org/CE0N/ontology/actor/ProducingResource</a>                                  |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                      |
| <b>Equivalentclass</b> | <a href="#">actorODP:ResourceRelation<sup>C</sup></a> and <a href="#">actorODP:participantRole<sup>OP</sup></a> value <a href="#">producer<sup>C</sup></a> |

## Providing Resource<sup>C</sup>

|                        |                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/ProvidingResource">http://w3id.org/CE0N/ontology/actor/ProvidingResource</a>                                  |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                      |
| <b>Equivalentclass</b> | <a href="#">actorODP:ResourceRelation<sup>C</sup></a> and <a href="#">actorODP:participantRole<sup>OP</sup></a> value <a href="#">provider<sup>C</sup></a> |

## Reselling Resource<sup>C</sup>

|                        |                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/ResellingResource">http://w3id.org/CE0N/ontology/actor/ResellingResource</a>                                  |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                      |
| <b>Equivalentclass</b> | <a href="#">actorODP:participantRole<sup>op</sup></a> value <a href="#">reseller<sup>C</sup></a> and <a href="#">actorODP:ResourceRelation<sup>C</sup></a> |

## Selling Resource<sup>C</sup>

|                        |                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/SellingResource">http://w3id.org/CE0N/ontology/actor/SellingResource</a>                                    |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                    |
| <b>Equivalentclass</b> | <a href="#">actorODP:participantRole<sup>op</sup></a> value <a href="#">seller<sup>C</sup></a> and <a href="#">actorODP:ResourceRelation<sup>C</sup></a> |

## Supplying Resource<sup>C</sup>

|                        |                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/SupplyingResource">http://w3id.org/CE0N/ontology/actor/SupplyingResource</a>                                  |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                      |
| <b>Equivalentclass</b> | <a href="#">actorODP:ResourceRelation<sup>C</sup></a> and <a href="#">actorODP:participantRole<sup>op</sup></a> value <a href="#">supplier<sup>C</sup></a> |

## Updating Resource<sup>C</sup>

|                        |                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/UpdatingResource">http://w3id.org/CE0N/ontology/actor/UpdatingResource</a>                                   |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                     |
| <b>Equivalentclass</b> | <a href="#">actorODP:participantRole<sup>op</sup></a> value <a href="#">updater<sup>C</sup></a> and <a href="#">actorODP:ResourceRelation<sup>C</sup></a> |

## Using Resource<sup>C</sup>

|                        |                                                                                                                                                        |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/UsingResource">http://w3id.org/CE0N/ontology/actor/UsingResource</a>                                      |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                  |
| <b>Equivalentclass</b> | <a href="#">actorODP:ResourceRelation<sup>C</sup></a> and <a href="#">actorODP:participantRole<sup>op</sup></a> value <a href="#">user<sup>C</sup></a> |

## Viewing Resource<sup>C</sup>

|                        |                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/ViewingResource">http://w3id.org/CE0N/ontology/actor/ViewingResource</a>                                    |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                    |
| <b>Equivalentclass</b> | <a href="#">actorODP:ResourceRelation<sup>C</sup></a> and <a href="#">actorODP:participantRole<sup>op</sup></a> value <a href="#">viewer<sup>C</sup></a> |

## Actor<sup>C</sup>

|            |                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------|
| <b>IRI</b> | <a href="http://w3id.org/CE0N/ontology/actor0DP/Actor">http://w3id.org/CE0N/ontology/actor0DP/Actor</a> |
|------------|---------------------------------------------------------------------------------------------------------|

## Capability<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/Capability>

**In Domain Of** [capabilityExtent<sup>op</sup>](#)  
[neededResourceRelation<sup>op</sup>](#)

## Collaboration Participation<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/CollaborationParticipation>

**Super Class Of** [CVNParticipation<sup>C</sup>](#)  
[ProcessParticipation<sup>C</sup>](#)

## Participation<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/Participation>

## Resource Relation<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/ResourceRelation>

**Sub Class Of** [actorODP:participantRole<sup>op</sup>](#) some [ActorResourceRole<sup>C</sup>](#) and [actorODP:participatingResource<sup>op</sup>](#) some [http://w3id.org/CEON/ontology/resourceODP/Resource<sup>C</sup>](http://w3id.org/CEON/ontology/resourceODP/Resource) and [actorODP:Participation<sup>C</sup>](#) and [actorODP:participatingActor<sup>op</sup>](#) some [actorODP:Actor<sup>C</sup>](#)

**Super Class Of** [BuyingResource<sup>C</sup>](#)  
[ConsumingResource<sup>C</sup>](#)  
[IssuingResource<sup>C</sup>](#)  
[OwningResource<sup>C</sup>](#)  
[ProducingResource<sup>C</sup>](#)  
[ProvidingResource<sup>C</sup>](#)  
[ResellingResource<sup>C</sup>](#)  
[SellingResource<sup>C</sup>](#)  
[SupplyingResource<sup>C</sup>](#)  
[UpdatingResource<sup>C</sup>](#)  
[UsingResource<sup>C</sup>](#)  
[ViewingResource<sup>C</sup>](#)

## Role<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/Role>

**Super Class Of** [ActorCollaborationRole<sup>C</sup>](#)  
[ActorResourceRole<sup>C</sup>](#)

## Cvn<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/cvn/CVN>

**Sub Class Of** <http://w3id.org/CEON/ontology/cvn/Collaboration><sup>C</sup>

## Collaboration<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/cvn/Collaboration>

**Super Class Of** <http://w3id.org/CEON/ontology/cvn/CVN><sup>C</sup>

## Process<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/processODP/Process>

## Resource<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/resourceODP/Resource>

## Object Properties

### capability extent<sup>op</sup>

**IRI** <http://w3id.org/CEON/ontology/actor/capabilityExtent>

**Sub Property Of** [actorODP:capabilityProperty](#)<sup>op</sup>

**Domain** [actorODP:Capability](#)<sup>C</sup>

### needed resource relation<sup>op</sup>

**IRI** <http://w3id.org/CEON/ontology/actor/neededResourceRelation>

**Sub Property Of** [actorODP:capabilityProperty](#)<sup>op</sup>

**Domain** [actorODP:Capability](#)<sup>C</sup>

### capability property<sup>op</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/capabilityProperty>

**Super Property Of**

- [capabilityExtent](#)<sup>op</sup>
- [neededResourceRelation](#)<sup>op</sup>

### participant role<sup>op</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/participantRole>

participating actor<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor0DP/participatingActor>

participating resource<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor0DP/participatingResource>

participation in<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor0DP/participationIn>

## Annotation Properties

contributor<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/contributor>

created<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/created>

creator<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/creator>

description<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/description>

license<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/license>

title<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/title>

preferred namespace prefix<sup>ap</sup>

**IRI** <http://purl.org/vocab/vann/preferredNamespacePrefix>

preferred namespace uri<sup>ap</sup>

**IRI** <http://purl.org/vocab/vann/preferredNamespaceUri>

covers requirements<sup>ap</sup>

**IRI** <http://www.ontologydesignpatterns.org/schemas/cannotation-schema.owl#coversRequirements>

## Namespaces

:

<http://w3id.org/CEON/ontology/actor/>

**actorODP**

<http://w3id.org/CEON/ontology/actorODP/>

**dcterms**

<http://purl.org/dc/terms/>

**odp**

<http://www.ontologydesignpatterns.org/schemas/cannotationschema.owl#>

**owl**

<http://www.w3.org/2002/07/owl#>

**prov**

<http://www.w3.org/ns/prov#>

**rdf**

<http://www.w3.org/1999/02/22-rdf-syntax-ns#>

**rdfs**

<http://www.w3.org/2000/01/rdf-schema#>

**vann**

<http://purl.org/vocab/vann/>

## Legend

|               |                          |
|---------------|--------------------------|
| <sup>c</sup>  | Classes                  |
| <sup>op</sup> | Object<br>Properties     |
| <sup>ap</sup> | Annotation<br>Properties |